CS471 – Lesson 13 – Reinforcement Learning
Value Iteration

In micro-blackjack, you repeatedly draw a card (with replacement) that is equally likely to be a 2, 3, or 4. You can either Draw or Stop if the total score of the cards you have drawn is less than 6. If your total score is 6 or higher, the game ends, and you receive a utility of 0. When you Stop, your utility is equal to your total score (up to 5), and the game ends. When you Draw, you receive no utility. There is a discount factor ($\gamma = 0.9$). Let's formulate this problem as an MDP with the following states: 0, 2, 3, 4, 5, and a Done state, for when the game ends.

1. Fill in the following table of value iteration values for the first 5 iterations, or if it converges only complete the table to convergence.

States	**0**	**2**	**3**	**4**	**5**
V_0					
V_1					
V_2					
V_3					
V_4					
V_5					

2. What is the optimal policy for this MDP?

States	**0**	**2**	**3**	**4**	**5**
π^*					